How to work

- Use the same `stata/do/` folder as in the previous weeks: the new data files are in `data/`, the worksheets in `exercises/`. In each worksheet, write the path of your own `stata/do` folder in the `cd` line.
- Part A of each worksheet contains the Guided Stata commands shown on the slides, in the same order. Part B is the laboratory: save the worksheet under a new name, complete the TODO lines and write your answers as comments.
- **Predict before you run**, and **read before you count**: every number produced by a dictionary is checked against the sentences it comes from.
- Laboratory work is formative and not graded.

Data for week 4

- `data/daycare_fines.dta`: 200 rows, one per day-care centre and week (10 centres × 20 weeks), transcribed by the instructor from Table 1 (p. 6) of Gneezy and Rustichini (2000). Variables: `centre`, `children`, `week`, `late` (late-coming parents in the week), `fine_group` (1 = centres 1–6), `period` (1 = weeks 1–4, 2 = weeks 5–16, 3 = weeks 17–20). As reported by the authors, week 11 had four days and its counts were multiplied by 5/4 (footnote 6); two centres were dropped because their recordings were incomplete (footnote 7). Source and changes: `data/SOURCE_AND_LICENCE_week4.txt`.
- `data/constitutions_text.dta`: 18,853 rows, one per sentence (or numbered clause) of the English text of 25 European constitutions, with `economy_code`, `country`, `constitute_id`, `year_adopted`, `year_text`, `sent_id` and `text`.
- `text_analysis/`, a folder to keep next to the folder `stata/`: the 25 original PDFs (`constitutions_pdf/`); the text files extracted from them, one sentence per line (`constitutions_txt/`); the extraction script shown in class (`08_pdf_to_txt.py`); and `LICENCE_AND_SOURCE.txt`. In Guided Stata 1 of lecture 8 you rebuild `constitutions_text.dta` from the text files.
- Source: Constitute Project (Elkins, Z., Ginsburg, T. and Melton, J., *Constitute: The World's Constitutions to Read, Search, and Compare*, constituteproject.org), licence CC BY-NC 3.0. Versions, changes (removed headings and page elements, merged fragments) and licence are listed in `text_analysis/LICENCE_AND_SOURCE.txt`.
- Two warnings. Most texts are translations, not the legal texts, and the translations have no legal force. Versions differ: each text includes amendments up to a different year (Italy: 2012).
- `data/justice_europe_2019.dta`: the European file of the previous weeks, used to compare the text measure with other indicators.

Lab 7: experiments

Question. Did the fine increase late arrivals at the day-care centres of Haifa, and how sure can we be with ten centres?

Worksheet. `exercises/07_experiments.do`.

Before class

Reread Sections I–II, Table 1 and Figure 1 of Gneezy and Rustichini (2000), and footnotes 6 and 7. Bring your draft comments (two or three) on the paper you are reviewing, written with the four steps.

Tasks

1. Compute the difference-in-differences for weeks 17–20 (after the fine was removed) against weeks 1–4, with standard errors clustered by centre. Is it larger or smaller than during the fine?
2. Estimate all three periods in one regression (`i.fine_group##i.period`). Which two coefficients reproduce the estimates of Part A and Task 1?
3. Placebo test: pretend the fine started in week 3 and use weeks 1–4 only. What do you expect the interaction to be, and what do you find? What does this say about parallel trends?
4. Measure late arrivals per 100 children and repeat the difference-in-differences of weeks 1–16. Why might a referee prefer this measure?
5. Randomization test by hand: list the change of each centre. Is every change in the fine group larger than every change in the control group? Explain why the probability of this pattern under "no effect" is 1/210 (`display comb(10,6)`).
6. Write one referee comment on the internal validity of the study, with the four steps; and a design for the text-message experiment of the slides (unit of randomisation, outcome, number of independent units, one check before looking at the results).

Self-check

- Did I count independent units (centres), not rows (centre-weeks)?
- Did I state the parallel-trends assumption and the evidence that supports it?
- Did I keep internal validity, external validity and ethics apart?

Key definitions

- Potential outcomes: each unit has an outcome with the treatment and one without; we observe only one. The effect is their difference; a control group stands in for the missing outcome.
- Random assignment: treated and control units are alike on average in everything, observed or not. With few units, the groups can still differ by chance.
- Difference-in-differences: (treated, after − before) − (control, after − before). In $y_{it} = \beta_0 + \beta_1 T_i + \beta_2 P_t + \beta_3 (T_i \times P_t) + u_{it}$, β_3 is the difference-in-differences (`i.T##i.P` in Stata).
- Parallel trends: without the treatment, treated units would have changed like the control units. Periods before the treatment are the evidence; a placebo test checks for an "effect" where there should be none.
- Clustered standard errors (`vce(cluster id)`): allow dependence within a unit observed many times. They count the units at which treatment was assigned; they can be larger or smaller than the usual ones. With few clusters, compare with an analysis that uses one number per unit.

- Randomization test: under "no effect", every assignment of the treatment is equally likely; the p-value is the share of possible assignments that give a result at least as extreme as the observed one.
- Internal validity: is the comparison credible for the units studied? External validity: would the result hold for other units, places, sizes of treatment or periods?

Lab 8: legal texts as data

Question. How do we turn a constitution into a number, and how do we know the number measures what we want?

Worksheet. `exercises/08_text_as_data.do`.

Before class: reading

Gentzkow, M., Kelly, B. and Taddy, M. (2019), Text as data, *Journal of Economic Literature* 57(3): 535–574, doi:10.1257/jel.20181020. Sections 1 and 2; Section 3.5.2 (model validation and interpretation). Skip the statistical methods in Section 3. Bring notes on:

1. What are the three steps that turn text into data? Which choices does each step involve?
2. What is a dictionary method, and why do the authors call it the most common method in the social sciences?
3. What kind of validation do they recommend for a dictionary measure?

Bring your draft comments on the paper you are reviewing: the second part of the lecture is feedback and revision in class.

Tasks

1. Audit 20 flagged sentences, drawn at random with the commands of Guided Stata 5. Code each sentence 1 if it is about the independence or impartiality of courts or judges, 0 if not. Compute precision. Which words misled the dictionary?
2. List the judiciary sentences of Denmark and of the Netherlands. Find the sentences that protect judges without using the words "independent" or "impartial", and quote them.
3. Build a second dictionary that adds *irremovable*, *subject only to*, *bound only by* and *solely by the law*. How many sentences are added? Are they correct? Which provision is still missed? (Hint: look for "only to the law" in Bulgaria.)
4. Replace judges with prosecutors. In how many constitutions does a sentence on prosecutors mention independence? Is every flagged sentence about the independence of prosecutors?
5. Compute the measure in two ways: flagged sentences as a share of judiciary sentences, and per 10,000 words. Do the two measures rank countries similarly? When would they differ?
6. Write one referee comment on a paper that uses a dictionary measure of legal texts, with the four steps; then revise your comments on the paper you are reviewing.

Self-check

- Did I convert the text to lower case before searching?

- Did I read a zero as "these words do not appear", not as "no protection"?
- Did I report precision together with the sentences I coded, and check whether the errors cluster in one country?
- Did I state which concept the measure captures: what is written, not what happens?

Key definitions

- From PDF to data: extract the text (dropping what is on the page only for the reader: headers, margin labels, headings), split it into documents (here sentences), and read one document per row.
- Text as data in three steps: represent the text as numbers (choose the document: here a sentence); map the numbers into a measure (here a dictionary); use the measure in an analysis.
- Dictionary: a list of words or phrases chosen by the researcher. A document is flagged if it contains them. Simple, transparent and reproducible; it reads words, not meaning.
- Regular expressions in Stata: `ustrregexm(text, pattern)` returns 1 if the pattern is found. In the pattern, `|` means "or" and `\b` marks the edge of a word. `ustrlower()` converts to lower case.
- Precision: the share of flagged documents that are correct. Recall: the share of correct documents that are flagged. Adding words usually raises recall and can lower precision; every new word must be audited again.
- Manual audit: compare what the dictionary flags with what a human reader would code, on a random sample of documents.
- Normalisation: documents differ in length, so counts must be compared with something (the number of relevant sentences, or the number of words). The choice can change the ranking.
- Referee checklist for text-based measures: which documents and versions; the unit of text; the dictionary and how it was chosen; the audit (precision); what is missed (recall); the concept the measure is meant to capture; the processing of the texts (extraction, translation, cleaning).